Offering “dip” promotes intake of a moderately-liked raw vegetable among preschoolers with genetic sensitivity to bitterness.

BACKGROUND: Evidence-based strategies for promoting vegetable consumption among children are limited. OBJECTIVE: To determine the effects of providing a palatable “dip” along with repeated exposure to a raw vegetable on preschoolers' liking and intake. PARTICIPANTS: One hundred fifty-two predominately Hispanic preschool-aged children studied in Head Start classrooms in 2008. DESIGN: A between-subjects, quasiexperimental design was used. A moderately-liked raw vegetable (broccoli) was offered twice weekly at afternoon snacks for 7 weeks. Classrooms were randomized to receive broccoli in one of four conditions differing in the provision of dip. Bitter taste sensitivity was assessed using 6-n-propylthiouracil. INTERVENTION: Broccoli was provided in four conditions: with regular salad dressing as a dip, with a light (reduced energy/fat) version of the dressing as a dip, mixed with the regular dressing as a sauce, or plain (without dressing). MAIN OUTCOME MEASURES: Mean broccoli intake during 7 weeks of exposure and broccoli liking following exposure. STATISTICAL ANALYSES: Descriptive statistics were generated. Multilevel models for repeated measures tested effects of condition and bitter sensitivity on mean broccoli intake during exposure and on pre- and post-exposure liking while adjusting for classroom effects and potential covariates. RESULTS: The majority of Hispanic preschoolers (70%) showed sensitivity to the bitter taste of 6-n-propylthiouracil. Children's broccoli liking increased following exposure but did not vary by dip condition or bitter sensitivity. Bitter-sensitive children, however, ate 80% more broccoli with dressing than when served plain ($P<0.001$); effects did vary based on whether regular or light dressing was provided as a dip or sauce. Dip did not promote broccoli intake among bitter-insensitive children. CONCLUSIONS: Providing dip—regular, light, or as a sauce—increased raw broccoli intake among bitter-sensitive Hispanic preschoolers. Findings suggest that offering low-fat dips can promote vegetable intake among some children who are sensitive to bitter tastes.